

MTE 203 – Advanced Calculus

Week 3 Homework

Note:

This Homework covers the material of Lectures 6, 7, 8 and 9 and it is therefore inherently long. You are encouraged to pace yourselves while working through the problems rather than attempting to complete everything in one sitting. Many of the questions involve graphing, which can be time-consuming and occasionally quite intricate, so please plan accordingly and start early.

Parametric and Vector Representations of Curves (S.11.10)

Problem 1 [11.10; Prob. 7]

Find the parametric and vector representation for the curve,

$$z = x + y$$

$$y = x^2$$

directed so that x increases along the curve. Draw the curve indicating its direction.

Problem 2 [11.10; Prob. 9]

Find the parametric and vector representation for the curve,

$$x = \sqrt{z}$$

$$z = y^2$$

directed away from the origin in the first octant. Draw the curve indicating its direction.

Differentiation and Integration of Vectors (S.11.9)

Problem 1 [11.9, Prob. 7, 19]

If $f(t) = t^2 + 3$, $g(t) = 2t^3 + 3t$, $\vec{u}(t) = t\hat{i} - t^2\hat{j} + 2t\hat{k}$, and $\vec{v}(t) = \hat{i} - 2t\hat{j} + 3t^2\hat{k}$, find the scalar or the components of the vector in the following exercises:

7: $\frac{d}{dt}[f(t)\vec{v}(t)]$

19: $\int [f(t)\vec{u} \cdot \vec{v}]dt$

Problem 2 [11.9, Prob. 25]:

Prove that if a differentiable function $\vec{v}(t)$ has constant length, then at any point at which $\frac{d\vec{v}}{dt} \neq 0$, the vector $\frac{d\vec{v}}{dt}$ is perpendicular to $\vec{v}$.

Displacement, Velocity and Acceleration (S.11.13)

Problem 1 [11.13, Prob. 17]:

A particle travels around the circle $x^2 + y^2 = 4$ counterclockwise at a constant speed, making 2 revolutions each second. If x and y are measured in meters, what is the velocity of the particle when it is at the point $(1, -\sqrt{3})$?

Problem 2 [11.13, Prob. 33]:

A boy stands on a cliff 50 m high that overlooks a river 85 m wide (see figure page 792 – Textbook Prob. 33). If he can throw a stone at 25 m/s, can he throw it across the river?

Length of Curves (S.11.11)

Problem 1 [S. 11.11, Prob. 11]:

Find the length of the following curve. Draw the curve.

$$x = 2 \cos t, \quad y = 2 \sin t, \quad z = 3t, \quad 0 \leq t \leq 2\pi$$

Problem 2 [S. 11.11, Prob. 13]:

Find the length of the following curve. Draw the curve.

$$x = t^3, \quad y = t^2, \quad z = t^3, \quad 0 \leq t \leq 1$$

Tangent Vectors (S.11.11)

Problem 1 [S. 11.11, Probs. 5]:

Express the following curve in vector form and find the unit tangent vector $\hat{\mathbf{T}}$ at each point in the curve:

$$x + y + z = 4, \quad x^2 + y^2 = 4, \quad y \geq 0$$

going from $(2,0,2)$ to $(-2,0,6)$

Suggestion: Sketch the curve for practice

Problem 2 [S. 11.11, Prob. 9]:

Find $\hat{\mathbf{T}}$ at the point $(5,2,10)$ of the curve $x = y^2 + 1$, $z = x + 5$, directed so that y increases along the curve.

Suggestion: Sketch the curve for practice

Curvature and Radius of Curvature (S.11.12)

Problem 1 [11.12, Probs. 15]:

Find the curvature and the radius of curvature of the curve (if it exists). Draw the curve.

$$x = t, \quad y = t^3, \quad z = t^2, \quad t \geq 0$$

Problem 2 [11.12, Probs. 17]:

Find the curvature and the radius of curvature of the curve (if it exists). Draw the curve.

$$x = t + 1, \quad y = t^2 - 1, \quad z = t + 1, \quad -\infty < t < \infty$$

Normal Vectors, Curvature and Radius of Curvature (S.11.12)

Problem 1 [11.12, Prob. 9] :

Find $\hat{\mathbf{N}}$ and $\hat{\mathbf{B}}$ at the point $(5,2,10)$ on the curve: $x = y^2 + 1$, $z = x + 5$, directed so that y increases along the curve.

Problem 2: [11.12, Prob. 23]

Let C be the curve $x = t, y = t^2$ in the xy -plane.

- a. At each point on C calculate the unit tangent $\hat{T}$ vector and the principal normal $\hat{N}$. What is $\hat{B}$?
- b. $\vec{F} = t^2 \hat{i} + t^4 \hat{j}$ is a vector that is defined at each point P on C . Denote by F_T and F_N the components of $\vec{F}$ in the directions $\hat{T}$ and $\hat{N}$ respectively. Find F_T and F_N as functions of t .
- c. Express F in terms of $\hat{T}$ and $\hat{N}$.

Extra Practice Problems¹

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.10, Probs. 6, 14
2. S. 11.9, Probs. 10, 18, 24
3. S. 11.11, Probs. 4, 10, 12, 14
4. S. 11.12, Probs. 10, 18, 24
5. S. 11.13, Probs. 12, 14, 24

¹ You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).